

Lab3 24b

Question 1

Write an application that uses objects to model the solar system. Create a class named Planet to encapsulate the name of a planet and the number of its moons. This contains a display method that outputs the encapsulated information (the planet name “ has ” so many “moons”).

Create another class named SolarSystem which holds references to the nine planets. (Mercury [0 moons], Venus [0 moons], Earth [1 moon], Mars [2 moons], Jupiter [16 moons], Saturn [18 moons], Uranus [15 moons], Neptune [8 moons], Pluto [1 moon]). An **array** is used in implementation. This also contains a display method that outputs the encapsulated information (using the array it outputs the information above – it uses the Planet output method).

Instantiate the **SolarSystem** class and invoke its **display()** method.

Question 2

You are required to develop the class IntegerSet. Each IntegerSet object can store integers in the range 0-100. The IntegerSet class uses an array of boolean values to record what integers are in the set. For example, array element a[i] is true if integer i is in the set, array element a[j] is false if integer j is not in the set. The no-argument constructor initializes the Java array to the “empty set” (i.e. a set whose array representation contains all false values).

Provide the following methods:

- Method union creates a third set that is the union of two existing sets (i.e. an element of the third set’s array is set to true if that element is true in either or both of the existing sets).
- Method intersection creates a third set that is the intersection of two existing sets (i.e. an element of the third set’s array is set to false if that element is false in either or both of the existing sets).
- Method insertElement inserts a new integer k into a set (by setting a[k] to true).

- Method `deleteElement` deletes integer `m` (by setting `a[m]` to false)
- Method `toString` returns a string containing a set as a list of numbers separated by spaces. Include only those elements that are present in the set. Use '---' to represent an empty set.
- Method `isEqualTo` determines whether two sets are equal.
- Method `isInSet` determines whether or not a specified integer is in the set. Returns a Boolean.

Write a program to test class `IntegerSet`. Test that all your methods work correctly.

Question 3

- 1) Using an array create the following `IntegerList` class.
- 2) Create an application (main class) to test your `IntegerList` class.

IntegerList
list: int [] total: int // counter control – used in add & remove
IntegerList(int) add(int): boolean remove(int): boolean isEmpty(): boolean isFull(): boolean getItem(int): int getTotal(): int